AI ASSISTED CODING

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Q1. Zero-Shot Prompting (Basic Lab Task)

Task: Write a Python function that classifies a given text as Spam or Not Spam using zero-shot prompting.

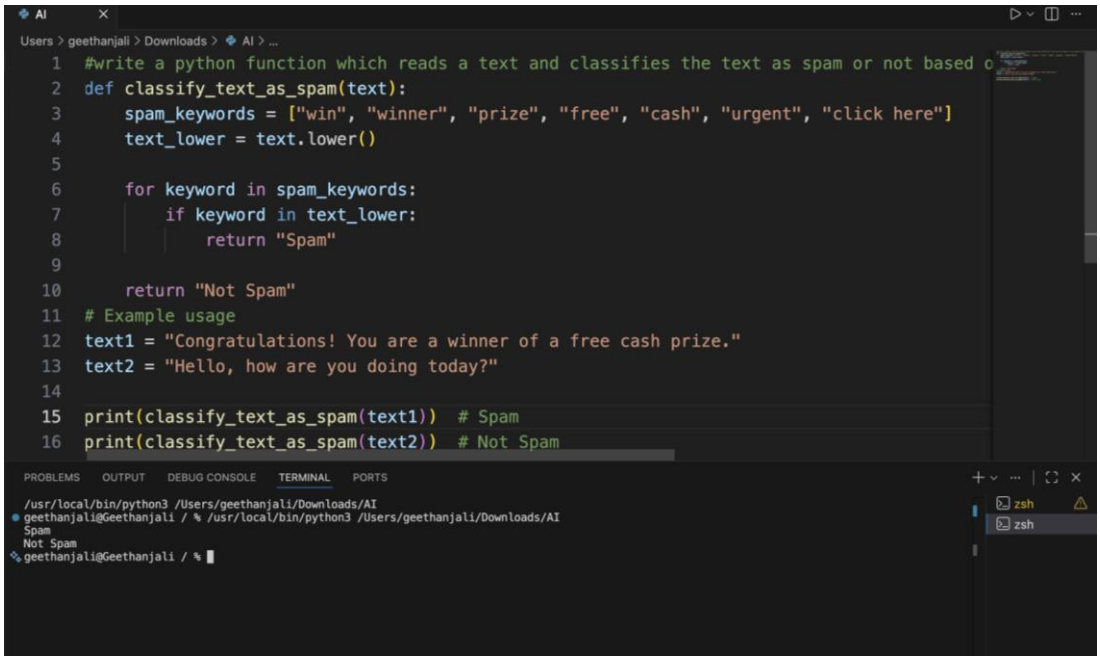
Steps:

1. Construct a prompt without any examples.
2. Clearly specify the output labels.
3. Display only the predicted label.

Input:

"Congratulations! You have won a free lottery ticket." Expected Output:

Spam



Q2. One-Shot Prompting (Emotion detection)

Task:

Write a Python program that detects the emotion of a sentence using one-shot prompting.

Emotions: [‘happy', 'sad', 'angry', 'excited', 'nervous', ’neutral’]

Steps:

1. Provide one labeled example inside the prompt.

2. Take a sentence as input.
3. Print the predicted emotion

Q3. Few-Shot Prompting (Student Grading Based on Marks)

Task:

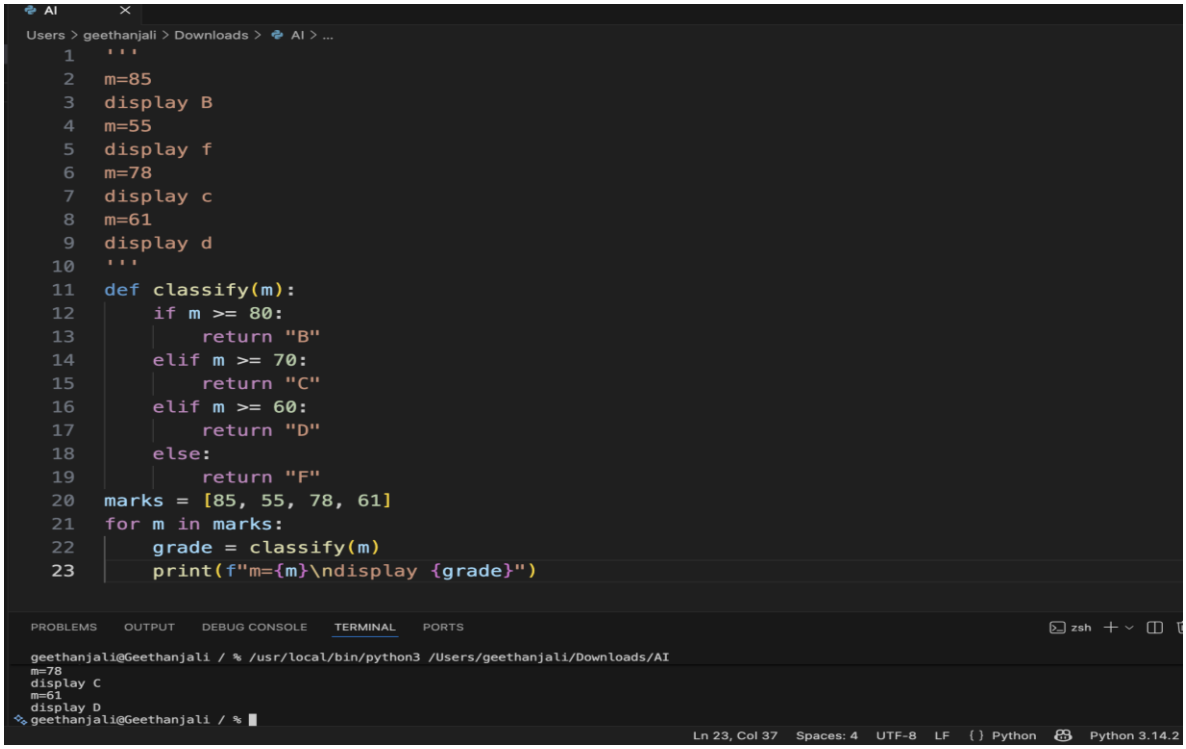
Write a Python program that predicts a student's grade based on marks using few-shot prompting.

Grades:

['A', 'B', 'C', 'D', 'F']

Grading Criteria (to be inferred from examples):

- 90–100 → A
- 80–89 → B
- 70–79 → C
- 60–69 → D
- Below 60 → F



Q4. Multi-Shot Prompting (Indian Zodiac Sign Prediction using Month Name)

Task:
Write a Python program that predicts a person's Indian Zodiac sign (Rashi) based on the month of birth (month name) using multi-shot prompting.

Indian Zodiac Order (Simplified Month-Based Model): The Indian Zodiac cycle starts in March with Mesha and follows this order: March → Mesha

April → Vrishabha

May → Mithuna

June → Karka

July → Simha

August → Kanya

September → Tula

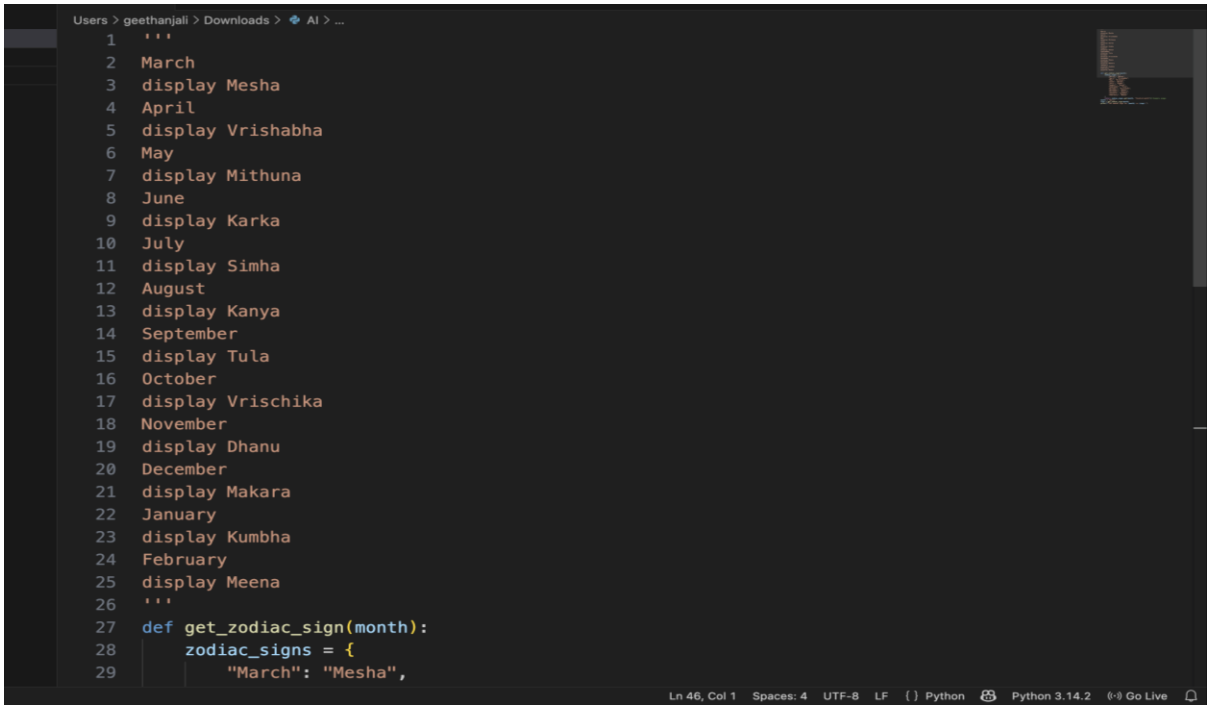
October → Vrischika

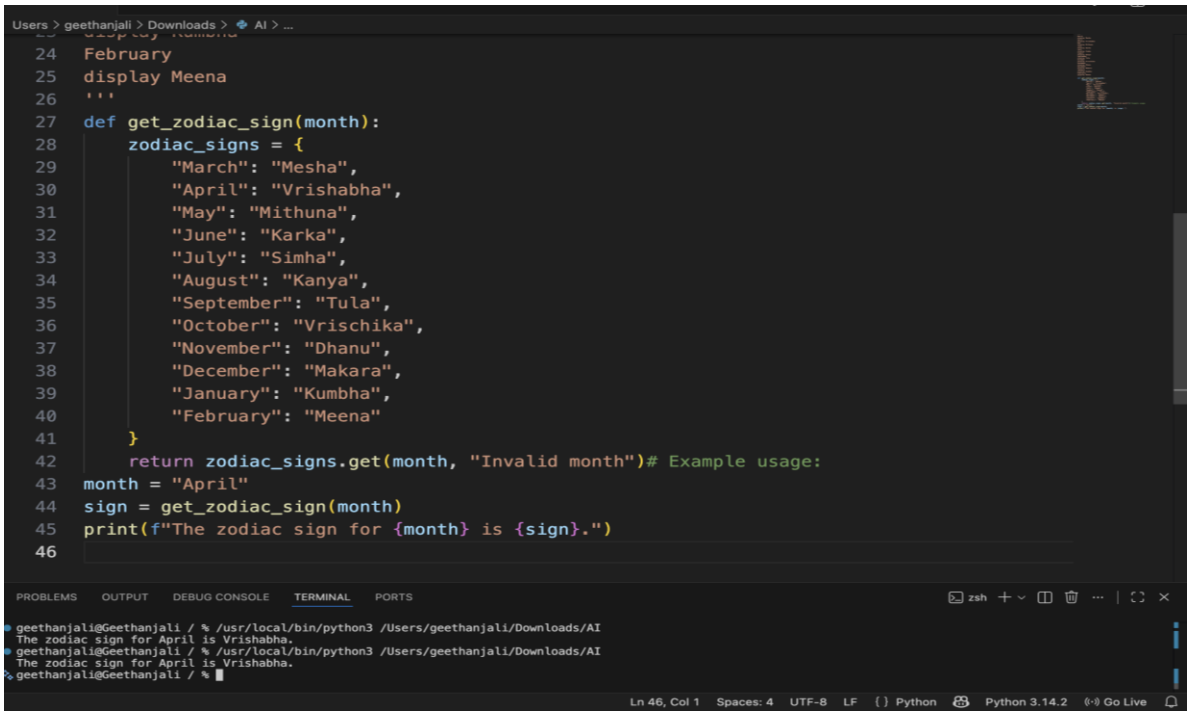
November → Dhanu

December → Makara

January → Kumbha

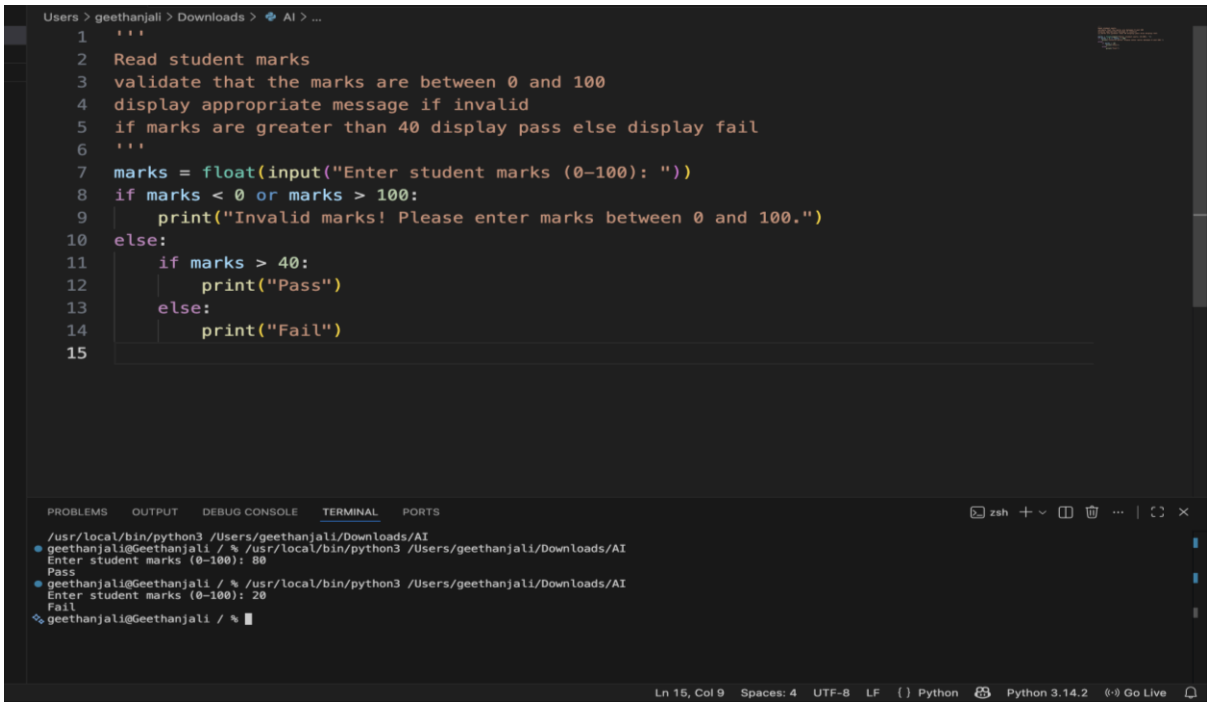
February → Meen


Users > geethanjali > Downloads > AI > ...
'''
March
display Mesha
April
display Vrishabha
May
display Mithuna
June
display Karka
July
display Simha
August
display Kanya
September
display Tula
October
display Vrischika
November
display Dhanu
December
display Makara
January
display Kumbha
February
display Meena
'''
def get_zodiac_sign(month):
zodiac_signs = {
"March": "Mesha",
Ln 46, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.2 Go Live


Users > geethanjali > Downloads > AI > ...
February
display Meena
'''
def get_zodiac_sign(month):
zodiac_signs = {
"March": "Mesha",
"April": "Vrishabha",
"May": "Mithuna",
"June": "Karka",
"July": "Simha",
"August": "Kanya",
"September": "Tula",
"October": "Vrischika",
"November": "Dhanu",
"December": "Makara",
"January": "Kumbha",
"February": "Meena"
}
return zodiac_signs.get(month, "Invalid month")# Example usage:
month = "April"
sign = get_zodiac_sign(month)
print(f"The zodiac sign for {month} is {sign}.")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
zsh
geethanjali@Geethanjali / % /usr/local/bin/python3 /Users/geethanjali/Downloads/AI
The zodiac sign for April is Vrishabha.
geethanjali@Geethanjali / % /usr/local/bin/python3 /Users/geethanjali/Downloads/AI
The zodiac sign for April is Vrishabha.
geethanjali@Geethanjali / %
Ln 46, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.2 Go Live

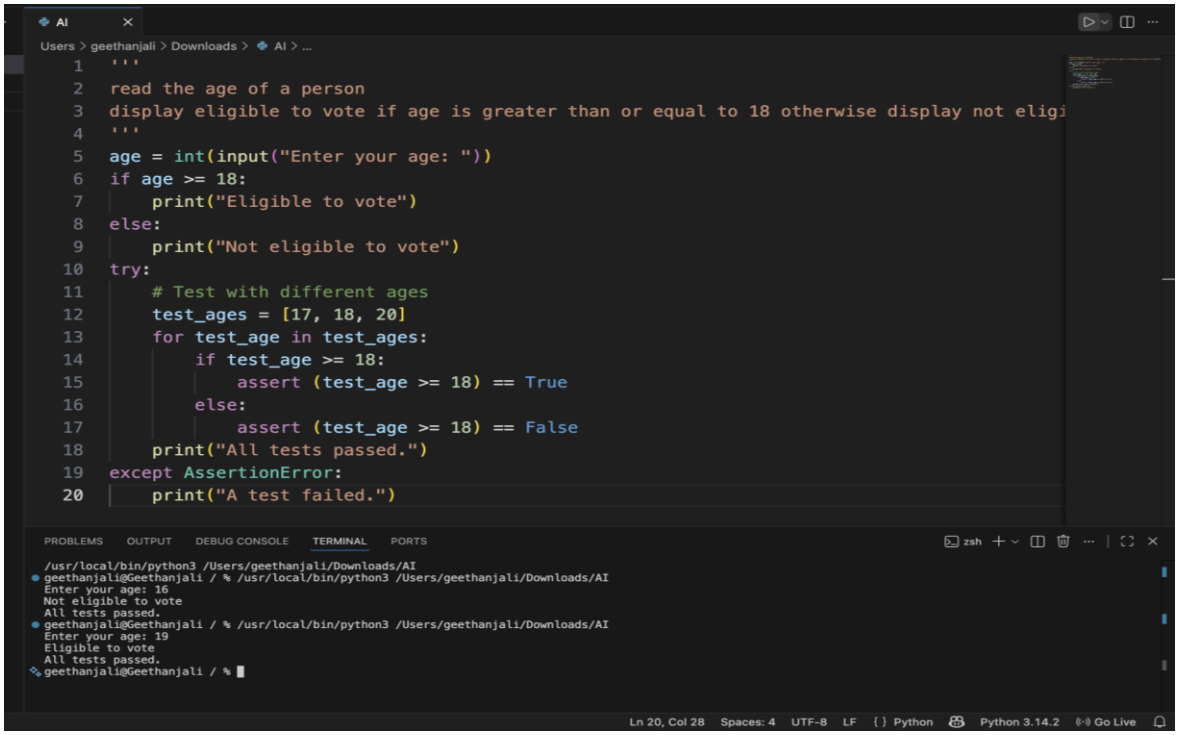
Q5. Result Analysis Based on Marks

Task: Write a Python program that determines whether a student Passes or Fails based on marks using Chain-of-Thought (CoT) prompting.
Result Categories: ['Pass', 'Fail']



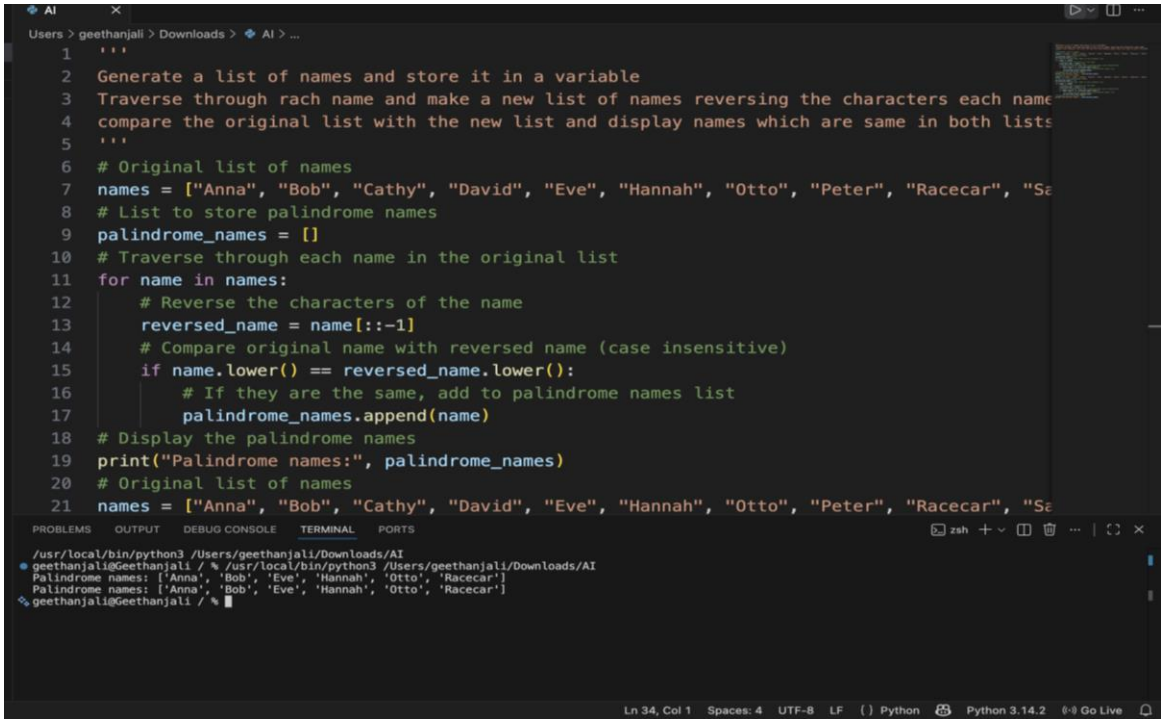
Q6 Voting Eligibility Check (Chain-of-Thought Prompting)

Task: Write a Python program that determines whether a person is eligible to vote using Chain-of-Thought (CoT) prompting.



Q7 Prompt Chaining (String Processing – Palindrome Names)

Task: Write a Python program that uses the prompt chaining technique to identify palindrome names from a list of student names.



Q8 Prompt Chaining (String Processing – Word Length Analysis)

Task: Write a Python program that uses prompt chaining to analyze a list of words. In the first prompt, generate a list of words.In the second prompt, traverse the list and calculate the length of each word. In the third prompt, use the output of the previous step to determine whether each word is Short (length less than 5) or Long (length greater than or equal to 5), and display the result for each word